

KINJAL SINGH

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EDUCATION: University of Illinois at Urbana-Champaign (UIUC)

Expected May 2026

Bachelor of Science in Computer Science and Statistics

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, JavaScript (Node.js, React), SQL, Rust, Bash

Full-Stack & Web: React, Node.js, Express, Flask, REST APIs, gRPC, HTML, CSS, TypeScript

Backend & Distributed Systems: Kafka, Redis, MongoDB, Hadoop, ELK Stack

Database Systems: PostgreSQL, DynamoDB, Redshift, Snowflake, BigQuery, Athena, Apache Spark, PySpark, Airflow, AWS Glue

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Helm, Terraform, GitHub Actions (CI/CD), Linux, Istio

Machine Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, MLflow, Hugging Face

WORK EXPERIENCE

Software Engineering Intern - Rivian

September 2025 – Present

- Designed and implemented a **Python-based benchmarking framework** to evaluate latency, throughput, memory bandwidth, and GPU utilization of steering-related perception and control workloads, improving validation throughput by **36%+**.
- Built **automation pipelines (Python, shell, pexpect)** to execute deterministic regression tests across hardware builds, validating steering sensor interfaces (UART, SPI, I2C, Ethernet) and SoC boot sequences.
- Profiled **C/C++ embedded** workloads supporting steering and chassis features to identify memory access inefficiencies, timing instability, and I/O bottlenecks.

Software Engineer Intern - PricewaterhouseCoopers (PwC)

July 2025 – August 2025

- Architected a self-healing **Agentic AI Ops platform** that analyzed real-time logs (**Kafka, ELK, Prometheus**) with **LSTM & Isolation Forest** models to detect and auto-remediate infra incidents.
- Deployed **predictive scaling** using **ARIMA & Prophet** integrated with **Kubernetes HPA** and **Terraform**, reducing downtime by **40%** and manual remediation by **60%**.
- Implemented **C++ gRPC microservices** for LLM inference, optimizing **ONNX runtime** and cutting latency **4×** vs Python-only deployments.

NLP Research Assistant- Respeaker Project at UIUC

February 2024 – May 2025

- Built an **edge software pipeline** with **Raspberry Pi** and **ReSpeaker**, enabling **real-time analytics** of classroom discussions through speaker diarization and engagement tracking.
- Engineered **Python** services with **Flask APIs** for diarization and sentiment tracking, integrating fine-tuned **DistilBERT & Wav2Vec2**, and achieved a model performance increase from **65% to 78% F1 score**.
- Optimized preprocessing and batch pipelines with **Python, Spark**, and **Airflow**, cutting end-to-end latency by **45%**, while scaling backend queries to **under 200ms** on **BigQuery** and **Athena** to support interactive classroom analytics.

Data Engineering Intern - Clover Infotech

May 2024 – July 2024

- Engineered distributed **ETL workflows** with **PySpark, Hadoop**, and **AWS Glue**, consolidating **5+ enterprise data sources** into **Snowflake and Redshift** for real-time analytics.
- Developed **Java based Lambda functions** for dynamic schema evolution & data validation, reducing ETL failures by **25%**.
- Built a **Python** and **React** service exposing ETL metadata as **REST APIs** for BI dashboards, enabling **1M daily API requests**.
- Improved pipeline observability with **Prometheus, Grafana** and **Linux-based log rotation strategies**, reducing incident response time by **38%**.

PROJECT HIGHLIGHTS

AutoLitigator – Full-Stack RAG Pipeline

January 2025 - March 2025

- Led a **3-member team** to design and deploy a **RAG backend (Pinecone and Redis)** handling **10K+ documents** with **sub-500ms** query latency.
- Built scalable microservices (**LangChain** and **FastAPI**) containerized on **Kubernetes (EKS)** with **HPA autoscaling**, sustaining **1K+ concurrent requests** at **95% uptime**.
- Implemented an **LLM-powered argument mining pipeline** that extracted statutes, precedents, and judge rationales from case texts, boosting legal query relevance by **35% in benchmark evaluations**.

CERTIFICATIONS

- AWS Certified:** Machine Learning – Specialty, Data Engineer – Associate, AI Practitioner, Cloud Practitioner
- Google Cloud Certified:** Professional Data Engineer, Cloud Generative AI Leader, Digital Leader

